

M.ABDULLAH IMRAN

ASSIGNMENT PART 4

PROGRAM#01

```
#include <iostream>

using namespace std;

int main() {

    int totalRows, totalCols;

    cout<<"Enter total rows and columns:"<<endl;

    cin >> totalRows >> totalCols;


    int marksTable[100][100];

    int totalSum = 0;

    for (int i = 0; i < totalRows; i++) {

        for (int j = 0; j < totalCols; j++) {

            cin >> marksTable[i][j];

            totalSum += marksTable[i][j];

        }

    }

    cout << totalSum << endl;

    return 0;

}
```

PROGRAM#02

```
#include <iostream>
```

```
using namespace std;

int main() {

    int totalRows, totalCols;

    // Input dimensions

    cin >> totalRows >> totalCols;

    int financialTable[100][100];

    for (int i = 0; i < totalRows; i++) {

        for (int j = 0; j < totalCols; j++) {

            cin >> financialTable[i][j];

        }

    }

    int subtractionResult = financialTable[0][0];

    for (int i = 0; i < totalRows; i++) {

        for (int j = 0; j < totalCols; j++) {

            if (i == 0 && j == 0) {

                continue;

            }

            subtractionResult -= financialTable[i][j];

        }

    }

    cout << subtractionResult << endl;

    return 0;
```

```
}
```

PROGRAM#03

```
#include <iostream>
```

```
using namespace std;
```

```
// Function to calculate area of a square
```

```
double calculateArea(double sideLength) {
```

```
    return sideLength * sideLength;
```

```
}
```

```
// Overloaded function to calculate area of a rectangle
```

```
double calculateArea(double rectLength, double rectWidth) {
```

```
    return rectLength * rectWidth;
```

```
}
```

```
int main() {
```

```
    double squareSide;
```

```
    double rectangleLength, rectangleWidth;
```

```
    // Input square dimensions
```

```
    cin >> squareSide;
```

```
    // Input rectangle dimensions
```

```
    cin >> rectangleLength >> rectangleWidth;
```

```
// Call overloaded functions and display results

cout << "Area of Square: " << calculateArea(squareSide) << endl;

cout << "Area of Rectangle: " << calculateArea(rectangleLength, rectangleWidth) << endl;

return 0;

}
```

PROGRAM#04

```
#include <iostream>

using namespace std;

double getBowlingAverage(int runsConceded, int wicketsTaken) {

    // Check to prevent division by zero crash

    if (wicketsTaken == 0) {

        return 0.0;

    }

    return (double)runsConceded / wicketsTaken;

}

int main() {

    int runs, wickets;

    // Input runs and wickets

    cin >> runs >> wickets;
```

```
double averageResult = getBowlingAverage(runs, wickets);

cout << "Bowling Average: " << averageResult << endl;

return 0;
}
```

PROGRAM#05

```
#include <iostream>

using namespace std;

void printWordVertically(char characterArray[], int arraySize) {

    for (int i = 0; i < arraySize; i++) {

        cout << characterArray[i] << endl;

    }

}

int main() {

    char userWord[100];

    int wordLength = 0;

    cin >> userWord;

    // Manually calculate the size/length of the character array

    while (userWord[wordLength] != '\0') {

        wordLength++;

    }

    printWordVertically(userWord, wordLength);

    return 0; }
```

PROGRAM#06

```
#include <iostream>

using namespace std;

// Function to reverse character array in place
void reverseWord(char characterArray[], int arraySize) {

    int startPoint = 0;

    int endPoint = arraySize - 1;

    while (startPoint < endPoint) {

        char temporaryHolder = characterArray[startPoint];

        characterArray[startPoint] = characterArray[endPoint];

        characterArray[endPoint] = temporaryHolder;

        startPoint++;

        endPoint--;

    }

}

int main() {

    char puzzleWord[100];

    int wordLength = 0;

    cin >> puzzleWord;

    while (puzzleWord[wordLength] != '\0') {

        wordLength++;

    }

    cout << "Reversed Word: " << puzzleWord << endl;

    return 0;}
```